

8(a)	<i>any two from</i> 1 <i>idea that</i> DELLA proteins prevent the activation of genes (for stem elongation) ; 2 gibberellin binds to receptors (on cell surface membrane) ; 3 causes breakdown of DELLA proteins ; 4 (so) transcription / gene expression / gene activation / mRNA synthesis, can occur ; 5 AVP ; e.g. <i>ref. to</i> transcription factors / PIF	2
8(b)	<i>any four from</i> 1 seed absorbs water ; 2 embryo produces gibberellin ; 3 gibberellin, moves to / acts on / stimulates, aleurone layer ; 4 (where) production of amylase occurs ; 5 amylase, hydrolyses / breaks down, starch in endosperm ; 6 to, maltose / glucose ; 7 embryo uses sugars for, respiration / growth ; 8 AVP ; e.g. gibberellins affect, gene / synthesis of mRNA, coding for amylase	4
8(c)	<i>any four from</i> 1 both P and Q same until, day 2 / paste applied ; 2 P, greater stem length / greater height / taller, than Q (after, day 2 / paste applied) ; 3 P 35 cm and Q 15 cm / P 20 cm longer than Q, at, end (of investigation) / day 20 ; 4 P greater rate than Q / AW ; 5 comparative calculated growth rates for P and Q ; e.g. 1.75 (cm day⁻¹) and 0.75 (cm day⁻¹) or 1.89 (cm day⁻¹) and 0.78 (cm day⁻¹)	4
8(d)	<i>any three from</i> 1 tall pea plants have, dominant allele / <i>Le</i> ; 2 (which codes for) enzyme that produces active gibberellin ; 3 dwarf pea plants (only) have, recessive alleles / <i>le</i> ; 4 (so) no (active) gibberellin formed ; 5 GA₁ is the active form of gibberellin ;	3